

ABSTRACT

Meetings are an essential part of organizations, educational institutions, software development teams, and business environments, where important information such as discussions, decisions, tasks, responsibilities, and deadlines are communicated. However, manually recording and reviewing meeting information is time-consuming and often results in the loss of important details. Participants may also find it difficult to concentrate on discussions while simultaneously taking notes. To overcome these challenges, this project proposes an **AI Meeting Assistant using Generative Artificial Intelligence**, an intelligent system designed to automatically process meeting recordings and generate meaningful and structured meeting information.

The proposed system combines **Speech-to-Text, Natural Language Processing (NLP), Generative AI, prompt engineering, embeddings, and Retrieval-Augmented Generation (RAG)** to provide an end-to-end solution for meeting management. The system accepts audio or video recordings of meetings as input. The recorded speech is converted into text using an AI-based speech recognition model such as **Whisper**. The generated transcript is then processed and analyzed using a Generative AI language model to understand the context and identify the most important information from the meeting.

The system automatically generates a **meeting summary, key discussion points, decisions, action items, responsibilities, deadlines, and open questions** from the transcript. Instead of presenting users with a lengthy transcript, the system provides concise and organized information that can be quickly reviewed. Prompt engineering techniques are used to guide the language model to produce consistent, relevant, and structured outputs while reducing unnecessary or unsupported information.

An important feature of the proposed system is **Retrieval-Augmented Generation (RAG)**, which enables users to search and ask questions about previously processed meetings. Meeting transcripts are divided into suitable sections and converted into numerical representations called embeddings. These embeddings are stored in a vector database. When a user asks a question, the system retrieves the most relevant meeting information and provides it as context to the Generative AI model. The model then generates an answer based on the retrieved meeting content. This allows users to ask questions such as *“What decisions were made in the previous meeting?”*, *“Who was assigned the testing task?”*, or *“What is the deadline discussed for the project?”*

The system follows a modular architecture consisting of a **React-based frontend, FastAPI backend, speech recognition module, Generative AI processing module, database, embedding service, vector database, and RAG-based search module**. The frontend provides an interactive dashboard through which users can upload meeting recordings, view AI-generated results, and search previous meetings. The backend manages file processing, transcription, AI analysis, database operations, and communication between different components. PostgreSQL can be used to store meeting metadata and generated results, while a vector database such as ChromaDB can be used for semantic retrieval.

The proposed AI Meeting Assistant improves meeting productivity by reducing manual note-taking and simplifying the process of reviewing long conversations. It provides a structured representation of meeting information and helps teams track important decisions and assigned tasks. The system can be particularly useful for project teams, corporate organizations, educational institutions, customer meetings, interviews, and online collaboration environments.

The project also demonstrates the practical application of **Generative AI in real-world information management**. By integrating speech recognition with language generation and RAG, the system goes beyond simple transcription and provides intelligent understanding of meeting content. Future enhancements can include **real-time meeting transcription, speaker identification, multilingual support, sentiment analysis, calendar integration, automatic email generation, task-management integration, personalized meeting insights, and cloud deployment**.

In conclusion, the **AI Meeting Assistant using Generative AI** provides an intelligent and efficient approach to meeting documentation and knowledge management. By automatically converting conversations into transcripts, summaries, decisions, and actionable tasks, and by enabling semantic search across previous meetings, the proposed system helps users save time, improve information accessibility, and make better use of meeting knowledge. The project demonstrates how modern Generative AI and NLP technologies can be integrated into a practical application to improve productivity and automate repetitive documentation tasks.

Keywords:-

Generative AI, AI Meeting Assistant, Speech-to-Text, Whisper, Natural Language Processing, Large Language Model, Prompt Engineering, Text Summarization, Action Item Extraction, Embeddings, Vector Database, Retrieval-Augmented Generation (RAG), FastAPI, React, PostgreSQL.